

AI Assisted Coding

Assignment-16.3

Name: B. Shravya

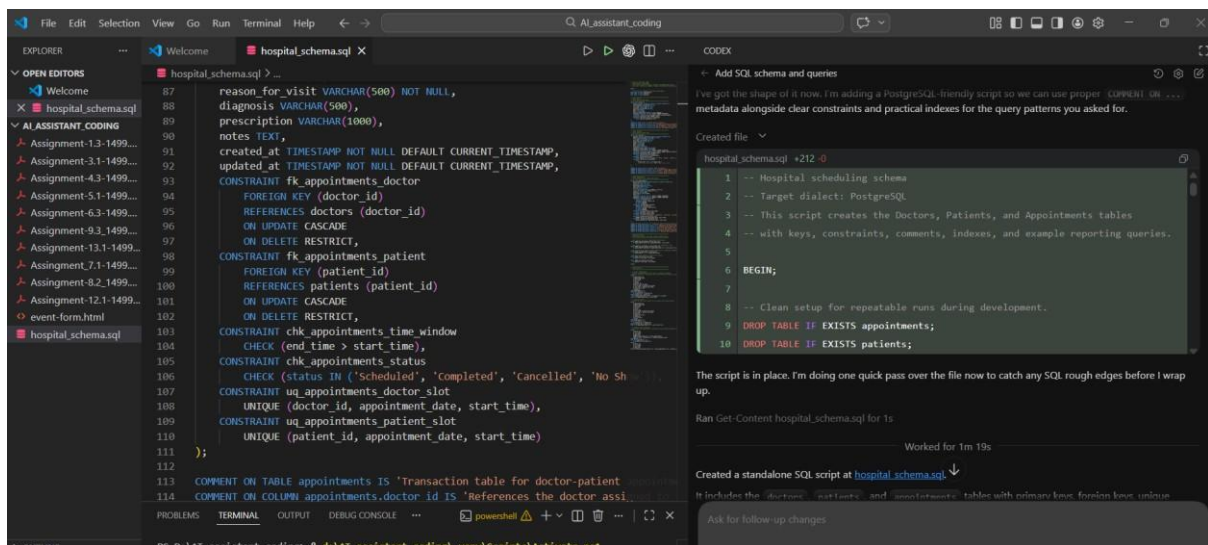
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Batch-08

Lab 16 – Database Design and Queries: Schema Design and SQL Generation

Task 1 – Hospital Management Database

Prompt: “Create SQL schema (Doctors, Patients, Appointments) with keys & constraints. Add queries: doctor appointments, patient history, patient count per doctor. Include comments + indexing.”

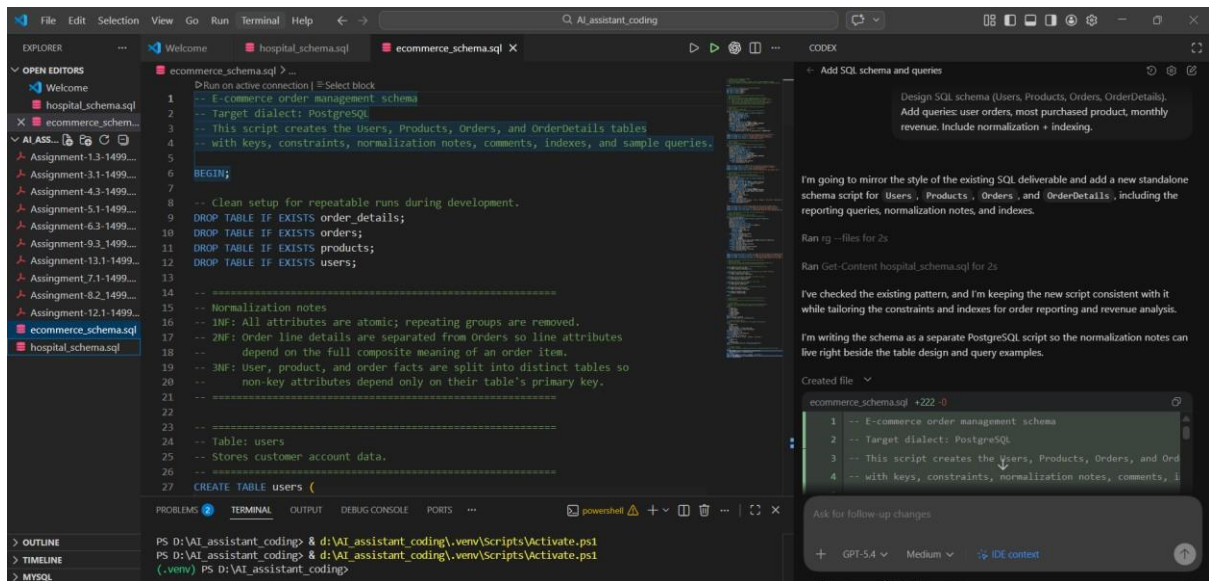


The screenshot shows the Visual Studio Code interface with the AI Assistant coding tool. The Explorer pane on the left shows the project structure with files like 'hospital_schema.sql' and 'event-form.html'. The main editor displays the SQL schema for the 'hospital_schema.sql' file, which includes tables for 'doctors', 'patients', and 'appointments' with their respective columns, constraints, and indexes. The AI Assistant pane on the right shows the generated SQL schema and queries, including comments and indexes. The terminal at the bottom shows the command to run the SQL script.

```
87 reason_for_visit VARCHAR(500) NOT NULL,
88 diagnosis VARCHAR(500),
89 prescription VARCHAR(1000),
90 notes TEXT,
91 created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
92 updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
93 CONSTRAINT fk_appointments_doctor
94 FOREIGN KEY (doctor_id)
95 REFERENCES doctors (doctor_id)
96 ON UPDATE CASCADE
97 ON DELETE RESTRICT,
98 CONSTRAINT fk_appointments_patient
99 FOREIGN KEY (patient_id)
100 REFERENCES patients (patient_id)
101 ON UPDATE CASCADE
102 ON DELETE RESTRICT,
103 CONSTRAINT chk_appointments_time_window
104 CHECK (end_time > start_time),
105 CONSTRAINT chk_appointments_status
106 CHECK (status IN ('Scheduled', 'Completed', 'Cancelled', 'No Show')),
107 CONSTRAINT uq_appointments_doctor_slot
108 UNIQUE (doctor_id, appointment_date, start_time),
109 CONSTRAINT uq_appointments_patient_slot
110 UNIQUE (patient_id, appointment_date, start_time)
111 );
112
113 COMMENT ON TABLE appointments IS 'Transaction table for doctor-patient
114 COMMENT ON COLUMN appointments.doctor_id IS 'References the doctor assi
115
116 -- Hospital scheduling schema
117 -- Target dialect: PostgreSQL
118 -- This script creates the Doctors, Patients, and Appointments tables
119 -- with keys, constraints, comments, indexes, and example reporting queries.
120
121 BEGIN;
122
123 -- Clean setup for repeatable runs during development.
124 DROP TABLE IF EXISTS appointments;
125 DROP TABLE IF EXISTS patients;
```

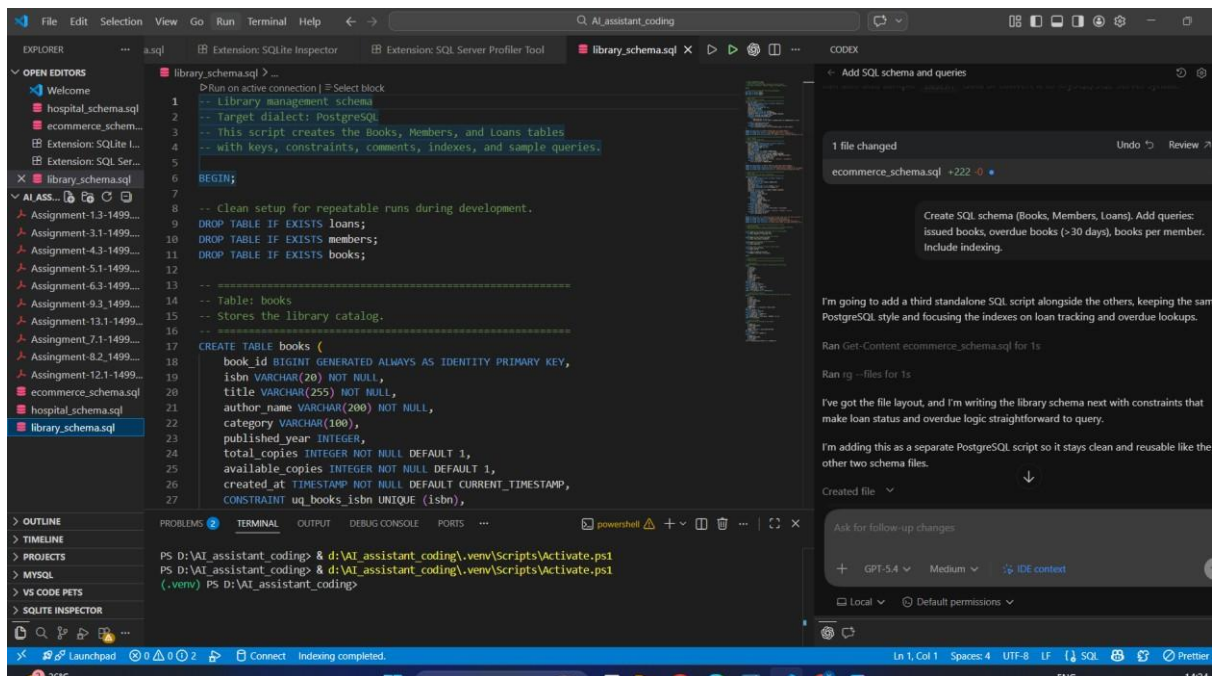
Task 2 – Real-Time Application: Online Shopping Database

Prompt: “Design SQL schema (Users, Products, Orders, OrderDetails). Add queries: user orders, most purchased product, monthly revenue. Include normalization + indexing.”



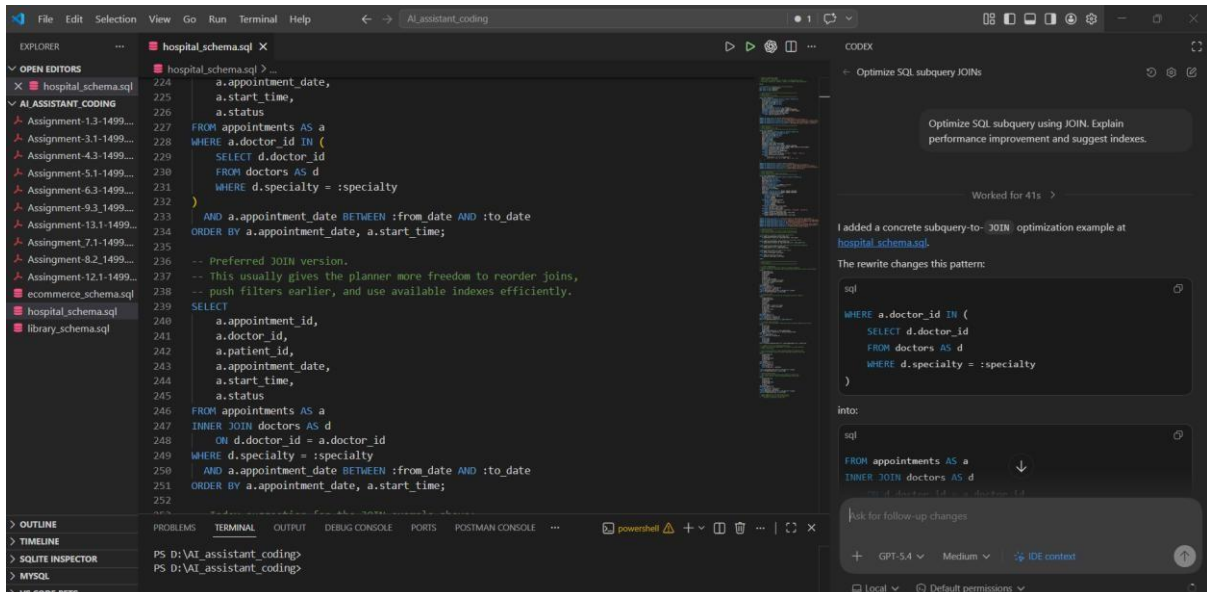
Task 3 – Library Database

Prompt: “Create SQL schema (Books, Members, Loans). Add queries: issued books, overdue books (>30 days), books per member. Include indexing.”



Task 4: Optimize Queries

Prompt: “Optimize SQL subquery using JOIN. Explain performance improvement and suggest indexes.”



Task 5: University Course Registration

Prompt: “Design SQL schema (Students, Courses, Faculty, Registrations). Add queries: students per course, faculty >2 courses, most registered course. Include optimization.”

